BTS – IV – 04.16 – 0480 Reg. No.

B

B.Tech. Degree IV Semester Examination April 2016

CS/IT 1405 DATA STRUCTURES AND ALGORITHMS
(2012 Scheme)

Time : 3 Hours Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) What is sparse matrix? Give its representation.
(b) Write an algorithm to perform insertion sort.
(c) Given two singly linked lists L1 and L2. Write an algorithm to merge two linked lists L1 and L2 into a single linked list.
(d) Explain deque.
(e) Is expression tree a binary tree? What is expression tree? Construct expression tree for the prefix expression *+ab-cd.
(f) What are the needs for creating threaded binary tree? What is two-way threading?
(g) Give array adjacency list representation of graph in memory.
(h) Explain Kruskal's algorithm to construct minimum cost spanning tree.

PART B

(4 × 15 = 60)

II. (a) Explain merge sort algorithm with example. Mention its complexity. (10)
(b) Write an algorithm to delete an element from an array. (5)

OR

III. (a) Write short notes on: (8)
(i) Hash table.
(ii) Hash function
(iii) Collision.
(iv) Linear probing.
(b) What is max heap? Explain heap sort algorithm with example. (7)

IV. Write an algorithm to translate an infix expression to postfix. Convert the infix expression $(A+B)\uparrow C-(D*E)/F$ to postfix. Note that $\uparrow$ is exponentiation operator. (15)

OR

V. (a) What are the advantages of circular queue over queue? Write an algorithm to insert an element into a circular queue and delete an element from a circular queue. (10)
(b) Differentiate singly linked list and doubly linked list. (5)

VI. (a) Develop a non-recursive algorithm to perform in order traversal of binary tree. (9)

(b) Explain AVL tree. (6)

OR

VII. What is binary search tree? How can insertion and deletion be done in a binary search tree? (15)

VIII. Explain graph traversal methods. (15)

OR

IX. (a) Write short notes on: $(5 \times 2 = 10)$

- (i) Undirected graph.
- (ii) Self edge.
- (iii) Spanning tree.
- (iv) Degree.
- (v) B-tree.

(b) Explain Dijkstra's algorithm, with an example. (5)

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